Markov Chain Monte Carlo (MCMC) Sampling - Part 2

ELG 5218 - Uncertainty Evaluation in Engineering Measurements and Machine Learning

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What is "wrong" with random-walk MH?

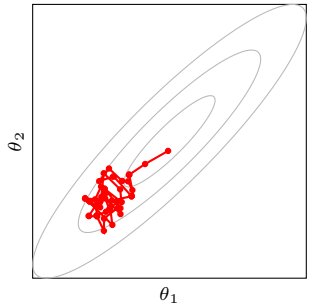
- Proposals are **blind**: $\theta^* = \theta + \epsilon$ ignores the shape of π.
- The chain moves by **small, undirected steps**; steps often undo each other.
  After n steps it has typically moved only $\propto \sqrt{n}$ step lengths.
- **Narrow, tilted valleys** (correlated parameters): the step must fit the narrow direction, so progress along the long direction is very slow.
- **High dimension**: the step size must shrink like $d^{-1/2}$ to keep a reasonable acceptance rate.

Bottom line

MH is correct, but it can be very inefficient. Idea of this part: use the **gradient** of $\log \pi$ to make better proposals.

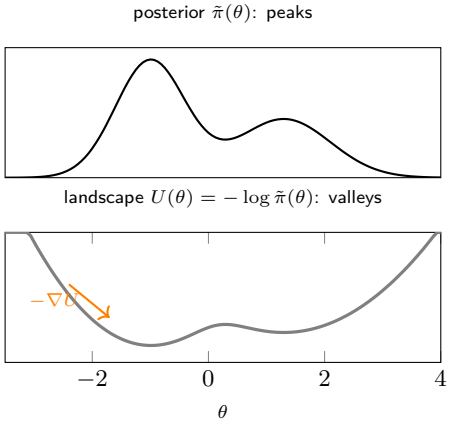
Interactive demos: `https://chi-feng.github.io/mcmc-demo/`

Random-walk MH, 60 iterations



Posterior with correlation 0.9. The chain jitters and has not explored the valley.

The landscape view: $U(\theta) = -\log \tilde{\pi}(\theta)$



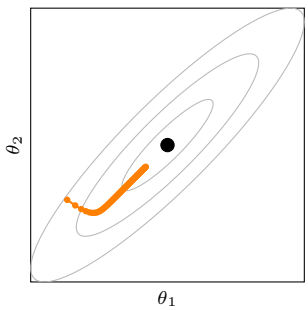
- High probability $\Leftrightarrow$ **low** U (a valley).
- $-\nabla U(\theta) = \nabla \log \tilde{\pi}(\theta)$ points **downhill**, toward higher probability.
- Only $\nabla \log \tilde{\pi}$ is needed: the evidence $p(x)$ is a constant and disappears from the gradient.
- For a posterior, $\nabla \log \tilde{\pi}(\theta) = \nabla \log p(x \mid \theta) + \nabla \log p(\theta)$, available by hand or by automatic differentiation.

Two physical pictures used below

Langevin: a particle in thick syrup on this landscape, randomly jostled.
HMC: a frictionless ball rolling on this landscape.

Why not simply follow the gradient?

Gradient steps only



- $\theta \leftarrow \theta - \eta \nabla U(\theta)$ is **optimization**: it ends at the peak (the MAP estimate) and stays there.
- All "samples" collapse onto one point: the spread, i.e. the **uncertainty**, is lost.
- A sampler must visit every region *in proportion to its probability*, including the tails.

What is needed

Gradient (go where probability is high) + **randomness** (do not get stuck at the peak) + a **correction** (exactly the right balance).

Langevin dynamics: drift plus noise

Physics: a particle in thick syrup on the landscape U. It slides downhill (drift) and is jostled by random molecular kicks (noise). It has no momentum: friction removes it immediately.

$$d\theta_t = -\tfrac{1}{2}\nabla U(\theta_t)\,dt + dW_t \qquad (W_t:\ \text{Brownian motion})$$

Why its long-run distribution is π

- Drift pushes probability **toward** the valleys.
- Noise spreads it **out** again.
- The two balance exactly when the density is $\pi \propto e^{-U}$ (the Fokker–Planck equation of this process has π as stationary solution).

Discretize with time step Δt

$$\tilde{\theta} = \theta_k - \tfrac{\Delta t}{2}\nabla U(\theta_k) + \sqrt{\Delta t}\,Z_k, \quad Z_k \sim \mathcal{N}(0, I)$$

- Accept every step: **ULA** (unadjusted). Simple, but slightly **biased**: discretization error.
- Add an MH accept/reject: **MALA**. **Exact** for any Δt.

MALA: Metropolis-adjusted Langevin algorithm

For $k = 0, 1, 2, \ldots$

1. **Propose** random number: $Z_k \sim \mathcal{N}(0, I)$
   $\tilde{\theta} = \mu(\theta_k) + \sqrt{\Delta t}\, Z_k, \qquad \mu(\theta) = \theta - \frac{\Delta t}{2} \nabla U(\theta)$ (drift toward higher probability)
2. **Accept** with probability random number: $u \sim \mathrm{Unif}(0, 1)$

$$\alpha = \min\left(1,\ \frac{\tilde{\pi}(\tilde{\theta})\ q(\theta_k \mid \tilde{\theta})}{\tilde{\pi}(\theta_k)\ q(\tilde{\theta} \mid \theta_k)}\right), \qquad q(a \mid b) \propto \exp\left(-\frac{\|a - \mu(b)\|^2}{2\Delta t}\right)$$

3. **Update:** $\theta_{k+1} = \tilde{\theta}$ if $u < \alpha$, otherwise $\theta_{k+1} = \theta_k$.

The proposal ratio cannot be dropped

The drift $\mu(\theta)$ depends on the starting point, so going $\theta \to \tilde{\theta}$ and $\tilde{\theta} \to \theta$ have different probabilities: q is **not symmetric**. Using only $e^{-U(\tilde{\theta}) + U(\theta_k)}$ samples the wrong distribution.

Choices: Δt tuned for acceptance ≈ 0.57 (vs. 0.234 for random walk). Cost: one gradient per iteration. In d dimensions the step can shrink like $d^{-1/3}$ instead of $d^{-1/2}$: fewer iterations for the same exploration.

Step 2.5: Bayesian logistic regression with MALA

Data: $X \in \mathbb{R}^{n \times d}$, $y \in \{0,1\}^n$ (0 = functional, 1 = failed). **Model:** $P(y_i = 1 \mid x_i, w) = \sigma(x_i^\top w)$, $w \sim \mathcal{N}(0, \lambda^{-1} I)$.

Landscape and its gradient

$$U(w) = -\sum_{i=1}^{n} \Big[y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i) \Big] + \tfrac{\lambda}{2} \|w\|^2$$

$$\nabla U(w) = -X^\top \Big(y - \sigma(Xw) \Big) + \lambda w$$

with $\sigma_i = \sigma(x_i^\top w)$. Cost $O(nd)$ per evaluation.

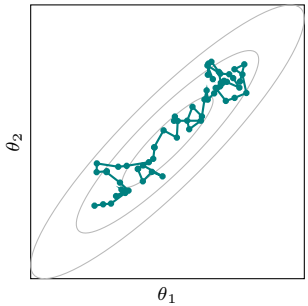
Why MALA here

- Random walk: $w^* = w + \epsilon$, blind.
- MALA: $w^* = w - \frac{\Delta t}{2} \nabla U(w) + \sqrt{\Delta t}\, Z$: the drift moves w toward weights that **explain the data better** (residuals $y - \sigma$ weighted by X), and the prior term pulls toward 0.
- Cost $O(nd)$ per iteration, like the likelihood in random-walk MH.

Remember the proposal ratio $q(w \mid w^*)/q(w^* \mid w)$ in the acceptance.

The limit of Langevin: momentum is thrown away

MALA, 60 iterations

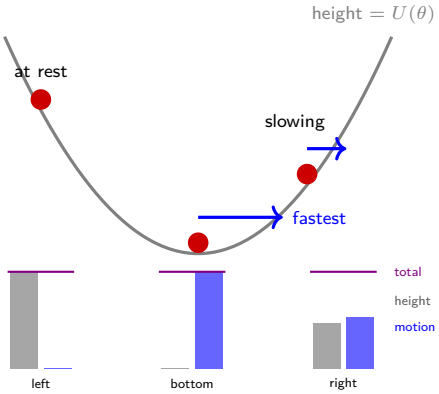


- MALA moves in good directions, but each step is small: the narrow direction of the valley limits Δt.
- Every step starts afresh: there is **no memory of the previous direction**. The motion is still diffusive (distance $\propto \sqrt{n}$).
- In the syrup picture: the particle never builds up speed.

Idea of HMC

Remove the friction. Give the particle a **momentum** and let it **roll** for many steps in a consistent direction before the next random kick.

HMC idea: give the parameter a velocity



- Imagine θ as a ball on the landscape $U(\theta) = -\log \tilde{\pi}(\theta)$.
- Besides its position θ, the ball has a **velocity**. In HMC it is called the **momentum** p (with mass 1 they are the same thing).
- Two simple rules:
  1. the velocity moves the position;
  2. the slope changes the velocity: downhill it speeds up, uphill it slows down.
- Without friction, **height energy** $U(\theta)$ and **motion energy** $K(p) = \frac{1}{2}\|p\|^2$ trade off; their **sum stays constant** (bars).

Consequence: the ball travels far through the valley, and it can climb only as high as its total energy allows, i.e. it stays where π is reasonably large.

Hamiltonian physics: the two rules in one function

The **state** is a point (θ, p) in *phase space*. One function, the total energy (the **Hamiltonian**), drives the motion:

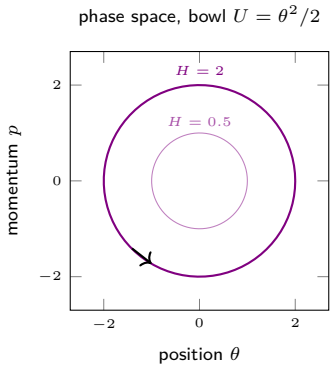
$$H(\theta, p) = \underbrace{U(\theta)}_{\text{height}} + \underbrace{\tfrac{1}{2}\|p\|^2}_{\text{motion}}.$$

Hamilton's equations are the two rules:

$$\frac{d\theta}{dt} = \frac{\partial H}{\partial p} = p \qquad \text{(velocity moves position)}$$

$$\frac{dp}{dt} = -\frac{\partial H}{\partial \theta} = -\nabla U(\theta) \qquad \text{(slope changes velocity)}$$

Property of the motion	**What it gives HMC**
energy H is conserved	end point as plausible as the start: (almost) always accepted
time-reversible ($p \to -p$ retraces)	symmetric proposal: no q ratio
phase-space volume preserved	no Jacobian correction
follows the valley	long, directed moves

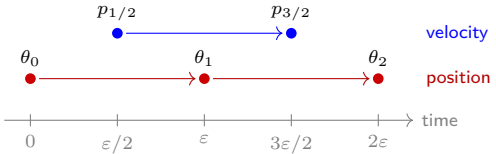


The ball swings back and forth; its state (θ, p) goes round a curve of constant H.

Leapfrog: computing the motion in small steps

The motion cannot be computed exactly for a real posterior, so we advance it in small time steps ε, like frames of a film. **Leapfrog** is the recipe for one frame:

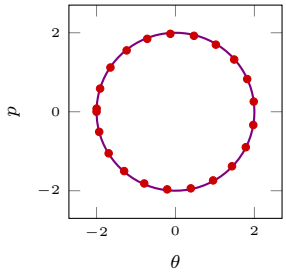
$$p \leftarrow p - \tfrac{\varepsilon}{2}\nabla U(\theta) \quad \text{half step of velocity (slope)}$$
$$\theta \leftarrow \theta + \varepsilon\, p \quad \text{full step of position}$$
$$p \leftarrow p - \tfrac{\varepsilon}{2}\nabla U(\theta) \quad \text{half step of velocity (new slope)}$$



Position and velocity are updated in turn and "jump over" each other, like children playing leapfrog.

- ε = time between frames: **accuracy**. L = number of frames; εL = how long the ball rolls: **distance**.
- One gradient per frame, so L gradients per iteration.

exact motion and leapfrog frames ($\varepsilon = 0.3$)



The frames stay close to the constant-energy curve: energy is almost conserved.

Why leapfrog

Energy almost conserved; **exactly reversible** and **volume-preserving**: the properties of the true motion that HMC needs.

HMC algorithm: where the random numbers are

From the current θ:

1. **Kick:** draw $p \sim \mathcal{N}(0, I)$. random
   Why: chooses a direction and speed, and *changes the energy level* (a new curve in phase space). Without it the ball would stay on one energy level forever.
2. **Roll:** L leapfrog steps of size ε from (θ, p). The **end point** (θ^*, p^*) is the proposal. deterministic
   Why: gradient-guided, long move along the valley. (NUTS will instead pick a random point of the path.)
3. **Accept/reject:** accept θ^* with probability $\min\left(1,\, e^{H(\theta,p)-H(\theta^*,p^*)}\right)$. random u
   Why: corrects the small leapfrog energy error. Exact dynamics would conserve H and always be accepted.
4. **Discard** p**.** The sample is θ only.

Langevin is HMC with $L = 1$

One leapfrog step with fresh $p \sim \mathcal{N}(0, I)$ gives $\theta^* = \theta - \frac{\varepsilon^2}{2}\nabla U(\theta) + \varepsilon p$: the MALA proposal with $\Delta t = \varepsilon^2$.

Step-size scaling with dimension d

random walk $d^{-1/2}$, MALA $d^{-1/3}$, HMC $d^{-1/4}$.
Optimal acceptance: 0.234, 0.57, 0.65.

Why does HMC sample from π?

Work with the joint density of position and momentum:

$$\pi(\theta, p) \propto e^{-H(\theta,p)} = \underbrace{e^{-U(\theta)}}_{\bar{\pi}(\theta)} \cdot \underbrace{e^{-\frac{1}{2}\|p\|^2}}_{\mathcal{N}(0,I)}.$$

θ and p are **independent** under this joint, and its θ-marginal is $\pi(\theta)$. So it suffices that both steps leave e^{-H} unchanged; then we keep θ and drop p.

Step 1: kick

Drawing a fresh $p \sim \mathcal{N}(0, I)$ is an exact draw from the conditional of p given θ (a Gibbs step). It leaves the joint unchanged.

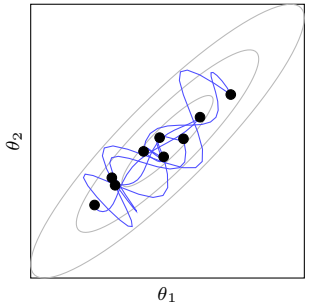
Steps 2–3: roll, accept

- Exact Hamiltonian flow keeps H (hence e^{-H}) constant and preserves volume.
- Leapfrog (followed by flipping $p \to -p$) is reversible and volume-preserving, so as a proposal it is **symmetric**: no q ratio.
- MH with ratio e^{-H^*}/e^{-H} fixes the small energy error.

The kick moves the chain *between* energy levels; the roll moves it far *along* one level. Together they explore the whole joint distribution.

Why do HMC paths curve, and why does that explore well?

HMC: 8 iterations, $\varepsilon = 0.15$, $L = 20$

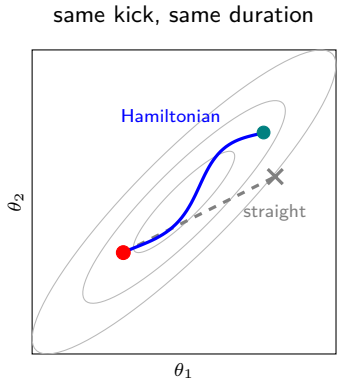


Blue: trajectories. Black dots: successive samples.

- The kick sets only the **initial** velocity. The slope keeps pushing the ball back toward the valley floor, so the path **bends**, like a ball thrown across a tilted bowl.
- The curvy look is not the goal in itself. The goal is that **successive samples are far apart** and nearly independent, *and* that the proposal is accepted.
- Along one trajectory the ball keeps its direction: distance grows $\propto$ time, not $\propto \sqrt{\text{steps}}$ as in a random walk.
- Energy is (nearly) conserved, so the end point lies in a reasonable region and is almost always accepted.

Compare with 60 random-walk or MALA iterations on the same valley. Each HMC iteration costs $L = 20$ gradients, so compare per gradient as well; the advantage grows with dimension.

Why this particular path? Compare with a straight line



Straight line (keep the initial velocity, ignore the slope): it leaves the valley. $\pi(\text{end})/\pi(\text{start}) = e^{-(4.9-0.74)} \approx 0.016$: almost surely rejected.

Hamiltonian path (the slope bends it back): it stays in the valley and travels as far. $H_{\text{end}} = 1.637 \approx H_{\text{start}} = 1.640$: accepted with probability ≈ 1.

So the path is chosen for three reasons

1. It **conserves energy**: far moves that are still accepted.
2. It is **reversible and volume-preserving**: the simple acceptance $e^{-\Delta H}$ is valid.
3. It **follows the shape** of π: no need to know the valley's direction in advance.

Tuning HMC: step size ε and trajectory length εL

	Too small	**Too large**
ε	accurate, but many gradients to go anywhere	energy error $\Rightarrow$ rejections; unstable above $\approx 2\sigma_{\min}$ (narrowest posterior SD)
εL	ball barely moves: random-walk behaviour returns	ball swings back toward its start (a **U-turn**): wasted gradients

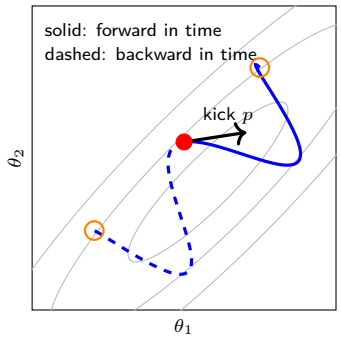
Physical rule of thumb

For a Gaussian direction with SD σ, U is a bowl and the ball swings with period $T = 2\pi\sigma$ (unit mass). Good trajectories last about **half a swing**; a full swing returns to the start. But σ differs across directions and across the posterior: no single L is right everywhere.

- ε is adapted during warm-up to reach a target acceptance (Stan default ≈ 0.8).
- L is the hard knob. **NUTS chooses it automatically, separately for every iteration.**

NUTS: grow the path both ways until it makes a U-turn

one NUTS trajectory



1. **Kick:** $p \sim \mathcal{N}(0, I)$. random
2. **Grow by doubling:** pick a time direction (forward/backward) random
   and add $1, 2, 4, 8, \ldots$ leapfrog steps to that end. deterministic
3. **Stop at a U-turn:** when the two ends (θ^-, p^-), (θ^+, p^+) start moving toward each other:

$$(\theta^+ - \theta^-) \cdot p^+ < 0 \quad \text{or} \quad (\theta^+ - \theta^-) \cdot p^- < 0.$$

4. **Pick the sample:** unlike plain HMC (end point), NUTS takes a **random point of the whole path**, with probability $\propto e^{-H}$ (nearly uniform). random

Backward in time = where the ball came from, i.e. the same physics started with $-p$. The red point is just one "frame" in the middle of the path.

Why is NUTS still exact? (intuition)

Problem with a naive stopping rule

"Roll forward from θ until the path turns" makes the path length depend on where you started. The starting point becomes special, the move is no longer reversible, and the chain samples the wrong distribution.

NUTS removes the special role of the starting point

- The current θ is treated as **one frame of a longer movie**. Growing in randomly chosen time directions builds the same path whichever of its frames you started from, with the same probability.
- Rules that would break this symmetry (e.g. a U-turn *inside* the newest doubling) cause that doubling to be discarded.
- The next sample is drawn among all points of the path with probability $\propto e^{-H(\theta,p)}$. Like a Gibbs step over "which frame", this leaves e^{-H}, and hence $\pi(\theta)$, unchanged.

Why not simply take the end point? Its position along the path would depend on the stopping rule and the starting point. Choosing randomly among all points keeps the symmetry. (Stan uses a multinomial variant that also favours the newest half of the path, to move farther; it stays exact.)

Where is the randomness? A summary

Method	Random	Deterministic	Correction
Random-walk MH	$\epsilon \sim \mathcal{N}(0, \sigma^2 I)$; accept u	–	$\tilde{\pi}$ ratio
MALA	$Z \sim \mathcal{N}(0, I)$; accept u	drift $-\frac{\Delta t}{2}\nabla U$	$\tilde{\pi}$ ratio $\times$ q ratio
HMC	kick p; accept u	L leapfrog steps; proposal = end point	$e^{-\Delta H}$
NUTS	kick p; doubling directions; which point is kept	leapfrog steps; U-turn check	built into the choice of point

Common pattern

Randomness to explore, the gradient to aim, and a correction (accept/reject or weighted choice) so that the chain spends time exactly in proportion to π.

Practical takeaways: when to use what

- **Random-walk MH:** no gradients needed; simple. Struggles with high dimension and strong correlation.
- **MALA:** one gradient per step; better directions, still diffusive. Useful when gradients are expensive; basis of stochastic-gradient Langevin for large data.
- **HMC:** long, gradient-guided moves; needs ε and L. (Implementations also adapt a "mass matrix" M, $p \sim \mathcal{N}(0, M)$, to rescale directions.)
- **NUTS:** HMC with L chosen automatically and ε adapted in warm-up; the default in Stan, PyMC, NumPyro. Requires continuous, differentiable parameters.
- Check diagnostics as before: $\hat{R}$, ESS, trace plots; for HMC/NUTS also **divergences** (the leapfrog simulation broke down in a region of high curvature).
- Strongly separated modes remain hard for all of them (tempering, multiple chains).

Interactive comparison (random walk, MALA, HMC, NUTS on 2D targets): https://chi-feng.github.io/mcmc-demo/